VitalQ — Personal Wellness & Longevity Coach

A retrieval-augmented AI assistant that synthesizes biomedical research, nutrition science, and behavioral data to deliver personalized, evidence-based wellness guidance.

Team	Group 1	**Members**	Juphens Cherfils · Kaden Glover · Richard Rodriguez · Yoana Cook
Course	ITAI 2377 · Data Science	**Approach**	Retrieval-Augmented Generation (RAG)
Institution	Houston Community College	**Due Date**	July 5, 2025
Instructor	Prof. Sitaram Ayyagari		

01 Executive Summary

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Purpose & Value Proposition

VitalQ is a domain-specific AI assistant designed to bridge the gap between cutting-edge longevity science and everyday health decision-making. Millions of individuals navigate complex, often contradictory health information without the context to act on it meaningfully. VitalQ addresses this by retrieving and synthesizing vetted biomedical literature, nutritional databases, and behavioral guidelines to provide personalized, source-grounded recommendations — without replacing clinical care.

The assistant targets health-conscious adults who value evidence over trends. Its core value lies in delivering the rigor of a clinical nutritionist and the breadth of a research librarian, accessible via natural language, available 24/7, and always cited.

RAG	5	3
Technical approach	Core capabilities	Primary data domains

02 Project Definition

20%

Domain Selection

Personal wellness and longevity — encompassing nutrition, sleep, metabolic health, movement, and stress physiology. Chosen for its rich public dataset ecosystem, high user demand, and direct alignment with RAG-based knowledge retrieval.

Problem Statement

Individuals seeking evidence-based health optimization face fragmented, low-quality information online. Existing tools lack citation transparency, personalization depth, or scientific rigor. VitalQ delivers traceable, research-grounded guidance.

Core Capabilities

1. Nutrition & micronutrient analysis · 2. Sleep quality optimization · 3. Biomarker interpretation · 4. Longevity protocol guidance · 5. Personalized daily health summaries

Target Users

Health-conscious adults aged 25–55, biohackers, preventive health adopters, and individuals managing chronic lifestyle conditions. Users value evidence over trends and seek actionable, personalized output.

Value Proposition

VitalQ delivers the rigor of a clinical nutritionist and the breadth of a research librarian — accessible via natural language, available 24/7, and always cited. It does not diagnose; it informs and empowers.

03 Data Requirements Analysis

25%

Data Types

Unstructured: PubMed biomedical abstracts, NIH clinical guidelines, WHO nutrition reports, peer-reviewed longevity research (PDFs/text).

Structured: USDA FoodData Central (nutritional values), CDC NHANES biometric datasets, user-entered health profiles (age, weight, goals, conditions).

Data Sources

PubMed / NCBI API · USDA FoodData Central · CDC NHANES · NIH Office of Dietary Supplements · WHO Global Health Observatory · Hugging Face Datasets

Volume & Velocity

Initial corpus: ~50,000 PubMed abstracts + ~300,000 USDA food entries. Refresh cadence: monthly for literature, quarterly for nutritional data. User profile data: real-time at query time. Total estimated vector store: ~2GB at launch.

Data Quality Requirements

Sources must be peer-reviewed or government-issued. Abstracts require DOI validation. Nutritional entries require completeness threshold (>80% fields populated). No anecdotal or commercial content admitted to the knowledge base.

Potential Data Challenges

1. **PubMed rate limits** — mitigated by batch API calls with exponential backoff and local caching.
2. **Contradictory research findings** — handled by retrieval confidence scoring and surfacing multiple perspectives.
3. **User data sparsity** — cold-start handled with demographic defaults and progressive profiling.
4. **Outdated guidelines** — publication date filtering applied at retrieval time (prefer publications within 5 years).

Data Schema

Table / Entity	Key Fields	Type
documents	doc_id, source, title, abstract_text, doi, pub_date, embedding_vector	**unstructured**
food_items	fdc_id, name, calories, protein_g, fat_g, carbs_g, fiber_g, micronutrients{}	**structured**
user_profiles	user_id, age, sex, weight_kg, health_goals[], conditions[], preferences{}	**structured**
query_logs	query_id, user_id, query_text, retrieved_doc_ids[], response_text, timestamp	**structured**
biomarkers	marker_id, name, unit, reference_range{low, high}, interpretation_text	**structured**

04 Processing Pipeline Design 25%

Data Preprocessing Workflow

1	**Data ingestion**	API pulls from PubMed (Entrez), USDA FoodData Central REST API, and CDC NHANES CSV files. Raw dat stored in staging directory. Duplicate detection via DOI hash for documents.
2	**Text normalization**	Lowercasing, punctuation stripping, medical abbreviation expansion (e.g., BMI to body mass index), removal HTML artifacts from PubMed XML. spaCy used for sentence segmentation.
3	**Structured data cleaning**	USDA entries: impute missing micronutrient values using category medians. NHANES: remove rows with >20% null fields. Standardize units (all weights to grams, all energy to kcal). Pandas + NumPy pipeline.
4	**Feature engineering**	Nutritional density score (nutrients per 100 kcal), biomarker risk flag (binary: within/outside reference range), document recency weight (decay function on pub_date), query-user relevance score (cosine similarity).
5	**Embedding generation**	sentence-transformers (all-MiniLM-L6-v2) applied to document chunks (512-token windows, 64-token overlap FAISS vector index built locally. User query embedded at runtime for similarity search.
6	**RAG retrieval & generation**	Top-k (k=5) documents retrieved per query. Augmented prompt constructed with user profile context + retrieved chunks. Response generated via LLM API. Citations surfaced inline from doc_id metadata.

Infrastructure

Google Colab (T4 GPU) for embedding generation. FAISS CPU index for retrieval (fits within free-tier memory). SQLite for structured data. No cloud deployment required for the plan-stage prototype.

Tech Stack

Python 3.10 · Pandas / NumPy · spaCy · sentence-transformers · FAISS · SQLite · Flask · Google Colab

05 Implementation Strategy

15%

Development Timeline

Week 1-2	**Data acquisition & schema setup** Pull PubMed corpus, USDA data, build SQLite schema. Assign data cleaning tasks.
Week 3-4	**Preprocessing pipeline** Text normalization, structured cleaning, feature engineering scripts completed in Colab.
Week 5-6	**Embedding & RAG build** Generate embeddings, build FAISS index, implement retrieval logic and prompt construction.
Week 7	**Flask API & interface** Wrap pipeline in REST endpoint. Build minimal query interface for demo.
Week 8	**Evaluation & final documentation** Run benchmarks, collect user feedback, finalize submission deliverables.

Team Member Responsibilities

JC

Juphens Cherfils

Executive summary & presentation

Section 1 · Final document assembly · Formatting & citations · Submission coordination

KG

Kaden Glover

Evaluation framework

Section 6 · Success metrics · Testing strategy · Performance benchmarks · User feedback design

RR

Richard Rodriguez

Domain research & project definition

Sections 1 & 2 · Literature review · Domain justification · Target user analysis

YC

Yoana Cook

Data architecture & pipeline lead

Data schema · Preprocessing pipeline · Feature engineering · Embedding generation · Flask API

06 Evaluation Framework

10%

>75%	<2s	100%	>4/5
Retrieval precision@5	Query response time	Responses with citations	User helpfulness rating

Testing Strategy	Continuous Improvement
Component-level: unit tests for each preprocessing step (null handling, normalization). Integration: end-to-end query tests with 50 curated health questions. Retrieval quality: manually annotated relevance judgments on 100 query-document pairs.	Query logs analyzed weekly for retrieval failures. User feedback (thumbs up/down per response) fed into a reranking layer. Literature corpus refreshed monthly via PubMed API. Embedding index rebuilt on significant corpus growth (>10%).

Statement of Contributions: All team members reviewed and approved this plan. Juphens Cherfils led executive summary and final document assembly. Kaden Glover led evaluation framework design. Richard Rodriguez led domain research and project definition. Yoana Cook led data architecture, schema design, processing pipeline, and technical implementation strategy. All members contributed to section review and final editing.